

Jean-Guillaume de Damas

📍 Paris, France ✉ jean-guillaume.de-damas@enpc.fr ☎ 07 70 70 67 75 🌐 jgdedamas.github.io
 🆔 0009-0008-6812-7530 🎓 Google Scholar 🔄 [jgdedamas](https://github.com/jgdedamas)

Experience

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| <p>ENPC - CERMICS, Postdoctoral Researcher</p> <ul style="list-style-type: none"> Working at the Centre d'Enseignement et de Recherche en Mathématiques et Calcul Scientifique on the development of hybrid model reduction methods Developing numerical methods for accelerating parametric studies in granular material simulations Implementing algorithms in the software SCOPi++ | <p>Paris, France
May 2026 – present
1 month</p> |
| <p>CERFACS, Master Intern - Researcher & Scientific Programmer</p> <ul style="list-style-type: none"> Researched randomized SVD preconditioners with applications to data assimilation Delivered functional C++ code for the European Weather Forecast Center Supervised by Selime Gürol | <p>Toulouse, France
Apr 2022 – Oct 2022
7 months</p> |

Education

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| <p>PhD INRIA Paris & Sorbonne Université, Applied Mathematics</p> <ul style="list-style-type: none"> Thesis: Randomized methods for solving eigenvalue problems and computing low-rank matrix approximations Supervised by Laura Grigori within the ERC project EMC2 | <p>Paris, France
Nov 2022 – Mar 2026</p> |
| <p>MSc ISAE-SUPAERO, Engineering</p> <ul style="list-style-type: none"> Machine learning: Algorithms for SVM, Neural Networks, Reinforcement learning and mathematical background Statistics: Bayesian inference, Stochastic algorithms, Non-parametric estimation High performance computing: Parallelization in C Optimisation: Inverse problem, data assimilation, problem regularization, and numerical challenges in inexact arithmetic | <p>Toulouse, France
Sept 2018 – June 2023</p> |

Publications

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| <p>Randomized Implicitly Restarted Arnoldi Method for the Non-Symmetric Eigenvalue Problem
 Vol. 46, No. 4, pp. 2395–2422. ISSN: 1095-7162.
 <i>Jean-Guillaume de Damas</i>, Laura Grigori
 10.1137/24m1674352 (SIAM Journal on Matrix Analysis and Applications)</p> | <p>Oct 2025</p> |
| <p>Randomized Krylov-Schur Eigensolver with Deflation
 <i>Jean-Guillaume de Damas</i>, Laura Grigori
 10.48550/ARXIV.2508.05400 (arXiv)</p> | <p>Jan 2025</p> |
| <p>Randomized Orthogonalization and Krylov Subspace Methods: Principles and Algorithms
 <i>Jean-Guillaume de Damas</i>, Laura Grigori, Igor Simunec, Edouard Timsit
 10.48550/arXiv.2512.15455 (arXiv)</p> | <p>Jan 2025</p> |

Test1

Programming: Julia, Python, Matlab, R, C, C++

Technologies: PyTorch, Scikit-learn, OpenMP, MPI, Cuda

Expertise: Numerical Analysis, Randomized Linear Algebra, Eigensolvers, Low-Rank Approximations

Test2

English: Advanced Level C1 (TOEFL ITP 600/677)

German: Intermediate (A2)

Japanese: Beginner (A1)

Test3

- Piano & Musicology: Learning as a passionate amateur with conservatory-level classes

Selected Talks

- Randomized Linear Algebra for the Eigenvalue Problem — SIAM Conference on Applied Linear Algebra (2024)
- Preconditioning for Data Assimilation with Randomized SVD — CERFACS Scientific Seminar (2022)